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A novel weld seam extraction method with semantic segmentation and point cloud feature for irregular structure workpieces

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ABSTRACT

The intricate surfaces of irregular structure workpieces present significant challenges for robotic welding path planning. To facilitate robot welding without teaching and programming, this paper proposes a weld seam extraction method that integrates semantic segmentation with point cloud features. This approach effectively harnesses the RGB-D information captured by an area array structured light camera. Initially, the K-Net model is employed for semantic segmentation of the workpiece using two-dimensional image features. This segmentation facilitates the coarse localization of the weld seam by extracting edges based on the segmented image mask, thus setting the stage for detailed weld seam extraction using point cloud features. Subsequent improvements in the DLP structured light vision imaging process allow for accurate edge point cloud reconstruction and adaptive extraction of ROI based on shape extension. The extraction of weld seam feature points is then performed using the LOBB feature extraction method, followed by a polynomial fitting of the weld seam using a least-squares approach. Experimental results indicate that the maximum error in weld seam extraction is less than 1.2 mm, with a root mean square error of less than 0.7 mm, and the algorithm completes its task in under 5 s, demonstrating the method's efficiency and precision.

1. Introduction

Irregular structure workpieces (abbreviated as ISW in this paper) typically exhibit complex contours and are often fabricated through methods such as stamping or casting. With the flourishing development of industries such as automotive manufacturing, consumer goods production, and aerospace, there is a sharp increase in demand for ISWs characterized by small batches, high variety, and complex structures. Traditional manual welding can no longer satisfy the processing requirements of ISWs in terms of efficiency and quality. Consequently, with the continuous maturation of robotic welding technology, the substitution of manual welding by robots has emerged as a crucial solution to meet these production demands [1,2].

One of the key technologies of welding robots is precise welding path planning. Traditional non-perceptive robot path planning methods mainly include teach-and-playback mode and offline programming [3,4]. These two types of methods excel in handling simple welding trajectories, enabling efficient and precise welding. However, traditional welding path planning methods may not be suitable for this type of workpiece because the complexity of its shape and structure makes path planning difficult. The added teaching and programming work

may compromise the efficiency and stability of production. Therefore, researching robotic welding path planning for ISWs without teaching and programming is of significant importance.

To address the limitations of traditional robot path programming, an increasing number of researchers are integrating vision sensors at the end of robot arms to form hand-eye systems. Through machine vision technology, these systems empower robots with perception capabilities, aiming to eliminate the need for complex robot path programming [5–7]. According to the sensors relied upon by the hand-eye system, welding path planning is mainly divided into two categories: those based on line structured light vision and those based on area array structured light vision.

Line structured light vision sensors project laser lines onto the target surface and capture the shape and position information of these laser lines through cameras to achieve the three-dimensional reconstruction and measurement of the target object. Due to its advantages, such as low cost, high precision, and fast measurement speed, it is highly favored by researchers [8–10]. For instance, to achieve tracking of narrow weld seams, Fan et al. enhanced the quality of laser image acquisition by adding light-emitting diodes to traditional line laser

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structures. Ultimately, they utilized a visual model to extract three-dimensional feature points of the weld seam [11]. Similarly, Cai et al. proposed a positioning method based on coordinate transformation and template matching using an improved cross-laser vision sensor. This method changed the traditional teach-and-playback method for tube plate welding [12]. Li et al. aimed to improve the robustness of weld seam feature point extraction by using Kalman filtering to limit the search scope. They extracted the weld seam by fitting centroids based on the least squares method [13]. Ma et al. proposed a sliding data queue method based on cubic B-spline fitting to overcome the issue of visual anticipation. This achieved efficient and accurate weld seam tracking for curved V-groove and lap weld seams [14].

The methods of welding path planning based on line structured light vision typically involve feature extraction and depth matching of two-dimensional images to obtain weld seam feature points, followed by online path correction and alignment [15,16]. Despite these advancements, line laser vision sensors, due to their limited scanning range, are primarily utilized for weld seam tracking and still necessitate rough teaching of welding paths, thus not yet achieving programming-free and teaching-free operations.

In contrast, area array structured light cameras project grid patterns with colored light onto the object surface using a projector. Then, the camera captures the scene illuminated by structured light to analyze and compute the three-dimensional surface information of the object. This type of camera only requires one shot to obtain complete three-dimensional information within its field of view [17,18]. Due to its high accuracy, strong anti-interference capability, and fast acquisition speed, it has found wide applications in welding scenarios. Unlike line structured light, welding robot path planning based on area array structured light mainly involves reconstructing the weld seam from the captured point cloud after obtaining the three-dimensional information of the workpiece, followed by offline path planning for welding. Therefore, this type of method enables welding robot trajectory planning without teaching and programming. Currently, these methods are mainly divided into two categories: those based on point cloud geometric fitting and those based on point cloud concave–convex feature.

In methods based on point cloud geometric fitting, these approaches segment the acquired workpiece point cloud through standard geometric fitting, and then solve for weld seam feature points based on the positional relationships of the weld seam [19]. For example, Geng et al. proposed a multi-plane fitting algorithm that achieved weld seam extraction for three types of medium-thick structures [20]. Lu et al. developed a novel Digital Light Processing (DLP) structured light camera by improving projection encoding. They utilized a RANSAC fitting algorithm to segment the workpiece surface from the obtained point cloud to extract weld seam feature points [21]. Zhang et al. initially classified lines and planes based on point cloud plane density, followed by straight line fitting of the obtained edge lines. Ultimately, they extracted weld seam feature points based on centroid positions [22].

In methods based on point cloud concave–convex features, these approaches describe features of the acquired point cloud. Point cloud features mainly include normals, curvature, and distribution of neighboring points, which describe the concavity and convexity of nearby points. Subsequently, feature selection and extraction are conducted in the feature space, ultimately achieving the extraction of weld seam feature points [23–25]. For example, Ahmed et al. extracted edges based on local neighborhood symmetry and identified corner points using curvature vectors. This method achieved edge and corner detection for welding workpieces [26]. Wang et al. proposed a weld seam detection algorithm based on point cloud voxel density for tube plate welding [27]. Patil et al. proposed a centroid translation method based on the feature distribution of edge points to select weld seams, enabling the extraction of spatially docked type I weld seams [28]. Gao et al. proposed a weld seam feature extraction algorithm based on point cloud normal boundaries for extracting type D weld seams [29]. Guo et al. utilized the Intrinsic Shape Signatures (ISS) keypoint detection

algorithm to extract weld seam Regions of Interest (ROI) and employed B-spline curve fitting for weld seam curve approximation [30]. Ma et al. performed non-rigid registration based on point cloud features between point clouds derived from CAD models and those collected on-site, achieving the conversion of the welding path extracted from the CAD model into the actual welding path [31].

However, there are still some issues with using the existing methods based on area array structured light vision to achieve weld seam feature extraction for the ISW:

1. Methods based on point cloud geometric fitting are typically only suitable for extracting weld seams from simple structured workpieces because these structures have relatively regular geometric shapes. However, the ISW often features complex free-form surfaces, rendering traditional geometric fitting methods unsuitable for accurately extracting weld seams.
2. Methods based on point cloud concave–convex features are applicable when the concave–convex features of the workpiece surface are primarily concentrated around the weld seam. However, for the ISW, the distribution of surface concavities and convexities is more scattered, making approaches relying solely on concave–convex features limited in accurately extracting weld seams.

Drawing inspiration from the literature, particularly the two-dimensional image processing used in line structured light vision methods, we propose an enhanced approach that improves upon the point cloud concave–convex feature-based approach by separately processing two-dimensional images and depth information to extract the weld seam features of the ISW [32]. Our main contributions are outlined as follows:

1. We propose a mask-based edge extraction method built on semantic segmentation, achieving weld seam coarse localization. This method demonstrates robustness by utilizing the segmentation mask to assist in guiding the edge detection process.
2. We develop an adaptive point cloud ROI extraction method based on shape extension, which optimizes the weld seam extraction process. This method dynamically adjusts to various surface geometries, ensuring precise weld seam feature extraction even under challenging conditions.
3. We leverage the concave–convex feature description of the point cloud to extract feature weld seam points and reconstruct the weld seam with high accuracy through curve fitting. This approach is particularly practical in handling irregular or complex weld profiles.

The structure of this paper is organized as follows: Section 2 introduces the construction of the welding robot based on 3D vision and outlines the overall process of the proposed method. Section 3 provides a detailed description of the weld seam feature extraction method based on semantic segmentation and point cloud features. Section 4 demonstrates the superiority of the proposed method through experiments. Section 5 summarizes the work presented in this paper.

2. System overview

2.1. System composition

According to Fig. 1, the robotic welding system based on 3D vision, constructed in this study, primarily consists of the following components: an industrial robot, a computer, an area array structured light camera, and welding equipment including a welding machine and a welding torch. For convenience in the narrative, the array structured light camera in the paper is referred to as a 3D camera.

Among these, the 3D camera and computer form the system's visual acquisition and processing section. Information about the workpiece is collected by the 3D camera and transmitted to the computer for

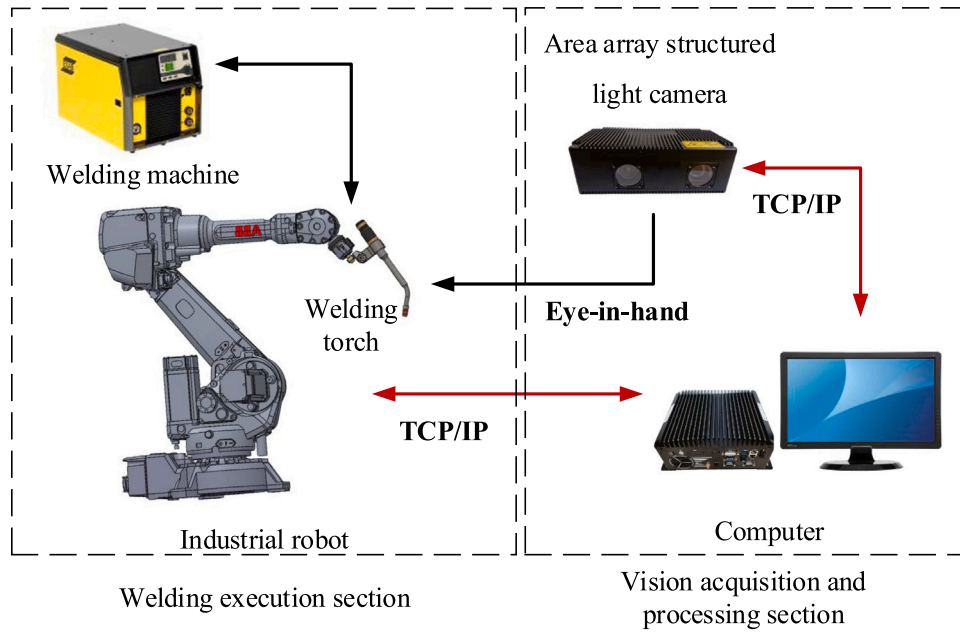


Fig. 1. Schematic diagram of the composition of the welding robot system based on 3D vision.

Table 1
Characteristic parameters of the 3D camera.

Case	Parameter	Value
1	Recommended working distance range (mm)	300–600
2	Near-end field of view (mm)	220 × 160 @ 0.3 m
3	Far-end field of view (mm)	430 × 320 @ 0.6 m
4	Resolution	1280 × 1024
5	Calibration precision (mm)	0.1
6	Typical acquisition time (s)	0.6–1.1
7	Overall dimensions (mm)	145 × 51 × 85
8	Weight (kg)	0.7

processing. The path planning algorithm designed in this study runs on the computer to process the data from the 3D camera. After taking into account the field of view, accuracy, size, and other factors of the 3D camera, we have ultimately chosen the MechMind Nano array structured light camera. The characteristic parameters of this 3D camera are shown in Table 1.

The industrial robot and welding equipment constitute the welding execution section, which is responsible for executing the welding paths generated by the path planning algorithm. Considering factors such as the precision, working range, and payload of the welding robot, we have selected the ABB IRB-2600 industrial robotic arm.

Data exchange and control command transmission between the system components are carried out through TCP/IP communication. The robot and 3D camera are combined into a hand-eye system with eye-in-hand.

2.2. Method framework

The proposed framework for the weld seam extraction method is illustrated in Fig. 2. This method is mainly divided into two stages: weld seam coarse localization and weld seam reconstruction.

In the weld seam coarse localization stage, the two-dimensional image from the ISW undergoes semantic segmentation using the K-net network first. From this, edge information is extracted based on the intersection locations of the masks obtained from semantic segmentation. Then, the depth image is combined with the edge information to reconstruct the edge point cloud. Finally, the ROI point cloud is obtained by extending the edge point cloud and serves as the input for

the weld seam reconstruction stage.

In the weld seam reconstruction stage, the ROI point cloud is first described using the LOBB feature to extract significant concave and convex weld seam feature points. Next, to further refine the accuracy, density information is utilized to filter the weld seam feature points. Finally, the weld seam is reconstructed by sampling the curve equation obtained from fitting the key points of the weld seam.

3. Method

3.1. Part segmentation based on deep learning

3.1.1. K-Net algorithm

To achieve semantic segmentation of the ISW, we have adopted a K-Net-based framework tailored for transfer learning. K-Net is a simple yet effective framework that unifies semantic, instance, and panoptic segmentation [33]. Specifically, it accurately segments instance and semantic class information through a set of dynamically updated convolutional kernels. Subsequently, it assigns learning objectives to each convolutional kernel using a mutually matching strategy and undergoes end-to-end training. This framework demonstrates high performance on mainstream datasets and meets the task requirements in this paper.

In semantic segmentation tasks, K-Net does not rely on specific model representation frameworks. Therefore, based on K-Net's excellent performance on the ADE20K semantic segmentation dataset, we chose the K-Net + UperNet method [34]. The framework architecture of this method is illustrated in Fig. 3. The network architecture uses Swin Transformer-Large (Swin-L) [35] as the backbone, extracting multi-scale features through hierarchical representations. The feature dimensions increase step by step from 192 to 1536 across different stages in this network. The backbone outputs are passed into the Feature Pyramid Network (FPN) for multi-resolution feature aggregation. A Pyramid Pooling Module (PPM) is applied to the highest-level features to capture global context. The fused multi-scale features from FPN and PPM form a feature map. Finally, it is processed by the Kernel Update Head. This head uses multi-head attention and linear layers to refine feature transformations, enhancing segmentation accuracy. It uses the Rectified Linear Unit (ReLU) activation function. By combining both local and global feature representations, the architecture ensures efficient and precise semantic segmentation.

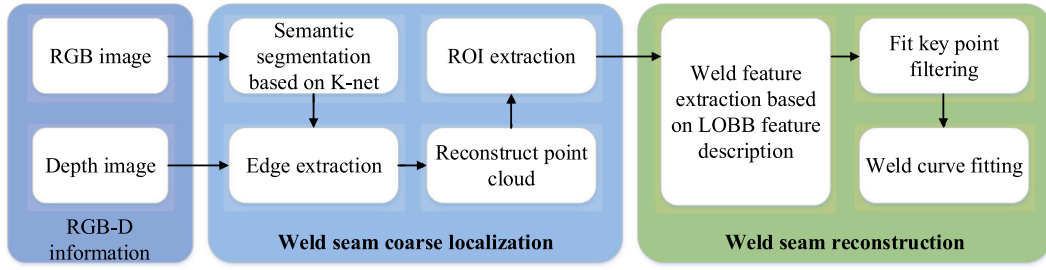


Fig. 2. Weld seam extraction method framework based on semantic segmentation and point cloud features.

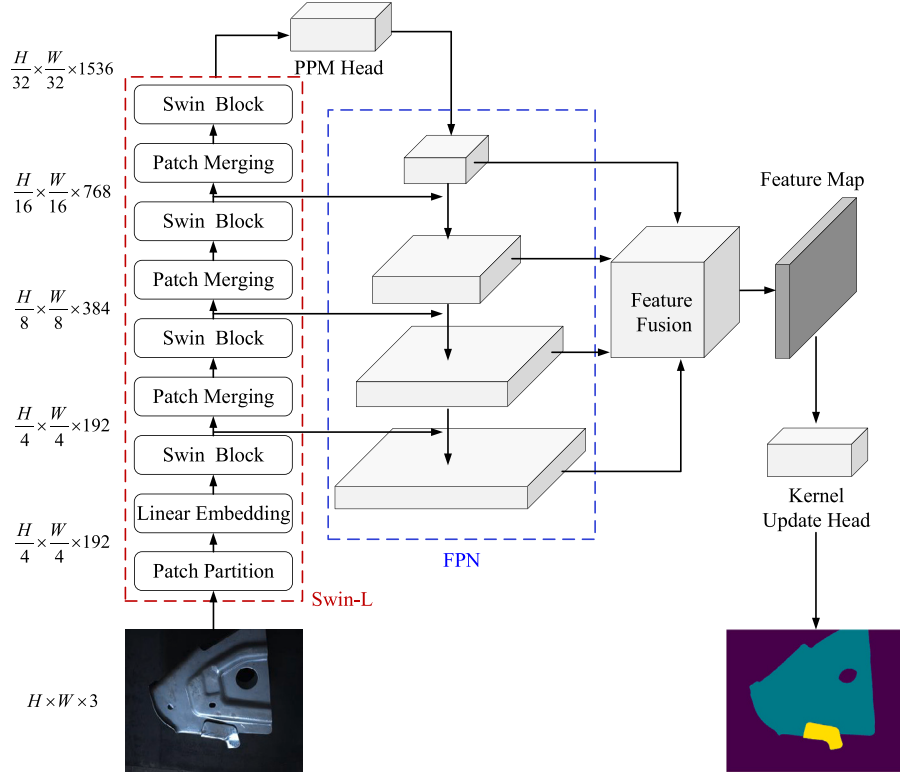


Fig. 3. Schematic diagram of K-Net + UperNet network architecture. The architecture mainly consists of a Swin-L backbone for multi-scale feature extraction, followed by an FPN for multi-resolution aggregation, a PPM for global context capture, and a Kernel Update Head that refines features.

To enable rapid network deployment, we used the MMSegmentation toolbox from OpenMMLab. This open-source framework offers a modular design and supports various state-of-the-art segmentation models and algorithms. It helped implement our model and streamline the training and evaluation processes. Further details can be found in the official documentation or codebase [36].

3.1.2. Dataset

The network used in this study was initially pre-trained on the ADE20K [37] semantic segmentation dataset, thus providing feasibility for transfer learning and reducing the demand for data. For this research, we collected 316 images to construct a semantic segmentation dataset for analyzing the ISW.

To ensure robust training, our dataset includes individual images of two distinct types of irregular-shaped components, as well as images that feature combinations of these components. These images encompass various perspectives and lighting conditions while maintaining data balance. Subsequently, we manually annotated the collected images, using “A” and “B” labels to differentiate between the two types of components. The annotated dataset was then divided into training

and validation sets in an 8:2 ratio. Some examples of the dataset are illustrated in Fig. 4.

To enhance the network’s generalization ability and prevent overfitting, we applied various augmentation techniques to the images during training. These included scaling, rotation, cropping, and adding noise.

3.1.3. Training

Based on the laboratory equipment, we set the Batch size to 2, the Initial learning rate to 0.01, the Learning momentum to 0.9, and the Epoch to 150 for network training. We used Stochastic Gradient Descent (SGD) for optimization. The model evaluation results during the training process are shown in Fig. 5. On the validation set, the metrics such as average accuracy (aACC), mean intersection over union (mIoU), and mean class accuracy (mACC) gradually improved, ultimately reaching 99.09%, 97.35%, and 98.75%, respectively. This indicates that the training process was effective and that the model exhibits strong generalization ability. On the training set, the model’s accuracy continued to improve throughout training, eventually reaching 99.31%. This excellent performance fully meets the requirements for workpiece segmentation discussed in this paper and lays a solid foundation for subsequent processing.

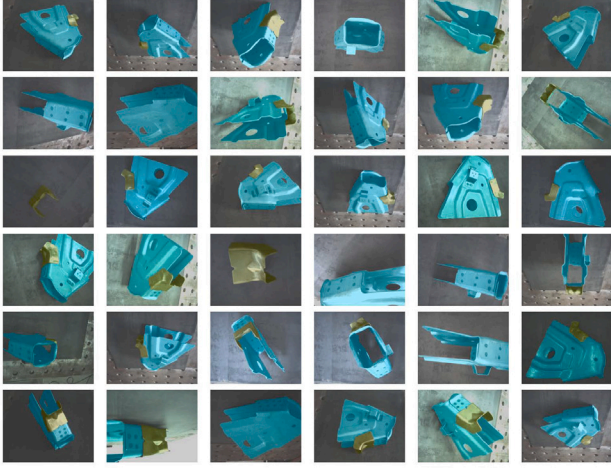


Fig. 4. A subset of images with corresponding annotations from the dataset. Blue represents A. Yellow represents B.

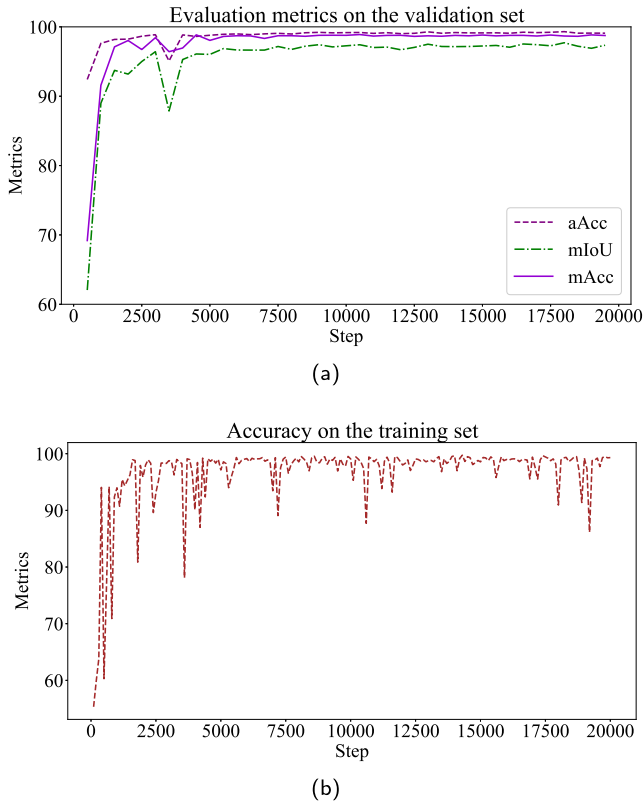


Fig. 5. Model evaluation line charts.

3.2. Weld seam coarse localization

Weld seams are typically located at the intersection of two irregularly shaped components. To coarsely localize the weld seam, the pixels where the components intersect are needed as a basis. First, based on the semantic segmentation results, we propose a mask-based edge extraction method. Then, we improve the imaging process of the DLP structured light camera to achieve point cloud reconstruction of the edge. An overview of the implementation process is shown in Fig. 6.

3.2.1. Edge extraction

In Section 3.1, we employed the K-Net network to obtain segmentation masks for the two distinct components. By analyzing the characteristics of the masks, we propose a mask-based edge extraction method.

As illustrated in Fig. 7, the intersection of the components can be viewed as a watershed of mask variations. Therefore, strictly defined edges are contours delineated by segmented pixels, which poses considerable challenges for precise extraction. Nonetheless, given that our objective is the coarse localization of weld seams, we adopt a generalized definition wherein these edges are represented as curves of pixels that exhibit a certain width. Based on this definition, we propose a method for discriminating edge pixels: When a pixel's neighborhood simultaneously contains two mask values, the pixel is considered an edge pixel. The specific solving process is as follows.

Let $P(x, y)$ denote the central pixel with coordinates (x, y) on the image, and the size of the neighborhood is $n \times n$ (where n is an odd number). This relationship is formalized in Eq. (1).

$$P(x, y) = \{M(i, j) | i, j \in [-2n - 1, 2n - 1]\} \quad (1)$$

Where $M(i, j)$ represents the pixel value in the neighbor area, i and j represent the horizontal and vertical offsets relative to the central pixel $P(x, y)$, respectively.

Then, the discrimination of edge pixels can be expressed as when there exists (i, j) satisfying the condition of Eq. (2), $P(x, y)$ is an edge pixel.

$$M(x + i, y + j) = mask1 \wedge M(x + i, y + j) = mask2 \quad (2)$$

Here, $mask1$ and $mask2$ are two different mask values.

3.2.2. Point cloud reconstruction

As shown in Fig. 6, a typical DLP structured light camera system consists of a projector and a camera. The projector utilizes DLP technology, often with the aid of LEDs or other light sources, to project structured light (such as stripes or grids) onto the surface of the target object. The camera passively captures images with phase shifts. Ultimately, through phase calculation techniques, it is possible to infer the three-dimensional shape information of the object's surface [38]. Additionally, the camera typically records unprojected two-dimensional images for texture mapping of the point cloud. In summary, the information obtained by the DLP structured light camera includes four channels: RGB-D, typically presented as a combination of a two-dimensional image and a depth image.

Each data point on the depth image can be transformed from its two-dimensional coordinates in the image coordinate system to three-dimensional coordinates in the camera coordinate system based on the camera's intrinsic parameters.

Assuming the camera's intrinsic parameters include u_0 and v_0 , which represent the column and row coordinates of the image center, respectively, and f_x and f_y are the focal lengths in the horizontal and vertical directions. Given a depth value at position (r, c) in the image coordinate system of the depth map as $depth$, the corresponding three-dimensional coordinates in the camera coordinate system are (x_c, y_c, z_c) . The transformation formula is as follows:

$$\begin{cases} x_c = \frac{depth \times (c - u_0)}{f_x} \\ y_c = \frac{depth \times (r - v_0)}{f_y} \\ z_c = depth \end{cases} \quad (3)$$

Based on the imaging principle of DLP structured light, both the two-dimensional image and the depth image share the same image coordinate system. Therefore, by integrating the RGB information at position (r, c) in the image coordinate system with the obtained three-dimensional coordinates, we can obtain point cloud data (x_c, y_c, z_c, r, c) .

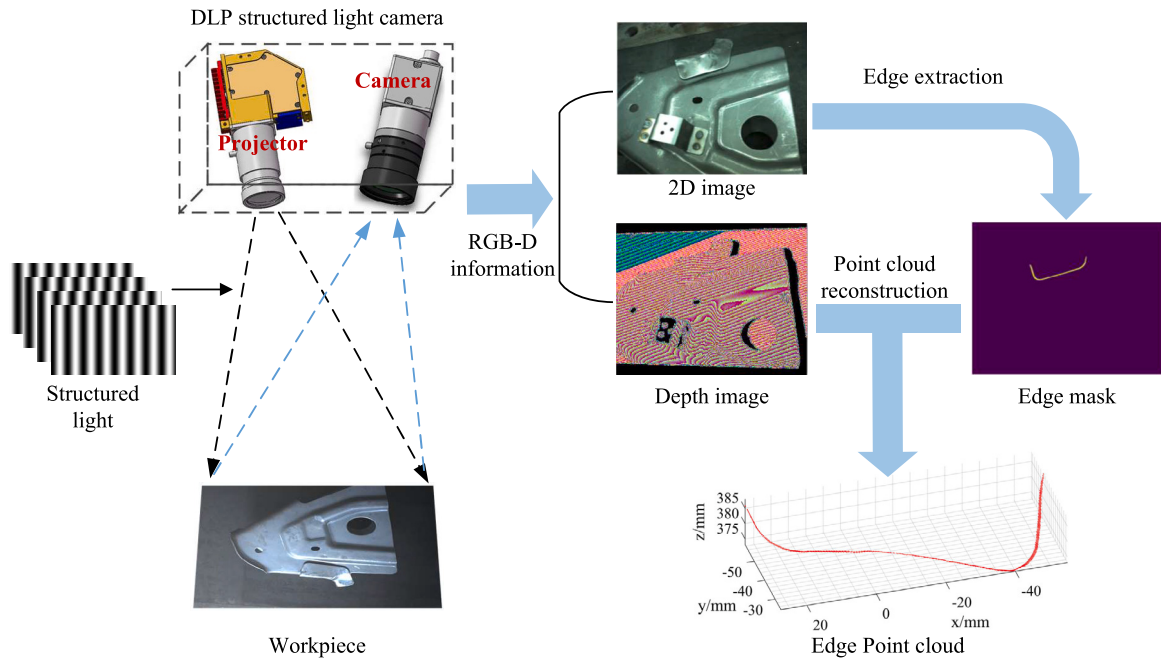


Fig. 6. The process of coarse localization of weld seams. The DLP structured light camera projects light and captures the RGB-D information of the workpiece. Edge extraction is performed based on the mask after semantic segmentation of the 2D image. This is then matched with the depth image to reconstruct the edge point cloud.

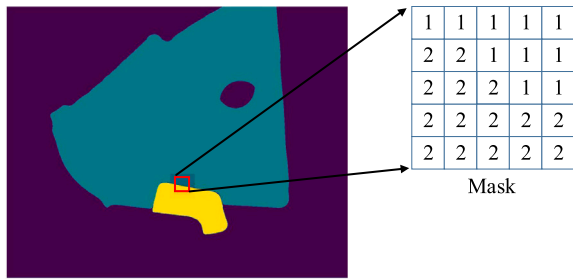


Fig. 7. Edge pixel mask diagram. 1 and 2 represent different mask values.

g, b). By traversing the depth image and substituting it into the reconstruction equation Eq. (3), we can reconstruct ordered point clouds with texture information.

As shown in Fig. 6, based on the point cloud reconstruction process described above, we can utilize texture information to filter the point cloud and obtain point clouds from specific regions. In Section 3.2.1, we perform edge extraction on the point cloud's two-dimensional image to generate a mask. Using the edge mask as a filtering criterion, we can obtain the point cloud from the edge region.

3.3. ROI extraction

Edge extraction based on two-dimensional features is susceptible to environmental lighting and image shadows. To enhance the accuracy and reliability of weld seam extraction, it is necessary to filter the point cloud around the weld seam based on three-dimensional features. Therefore, this paper uses the edge point cloud extracted from the process shown in Fig. 6 as the coarse localization of the weld seam. By extending the ROI based on this point cloud, a subset of the point cloud around the weld seam can be obtained. This serves as preparatory work for extracting weld seam feature points based on the concave-convex features of the point cloud.

Due to the complexity of the weld seam curves in ISWs, traditional ROI extraction methods tend to introduce noise, reducing the efficiency of weld seam feature extraction. To address this, we propose an

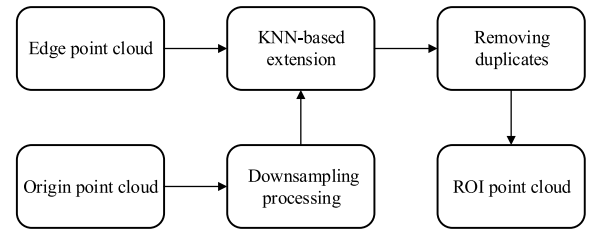


Fig. 8. Processing steps of the adaptive point cloud ROI extraction method based on shape extension.

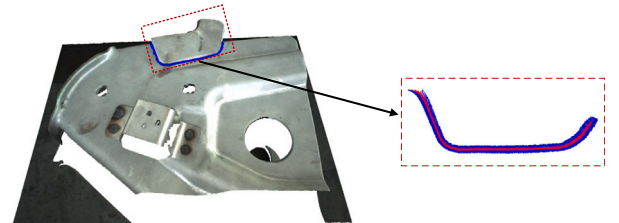


Fig. 9. ROI extraction result. Red represents the edge point cloud. Blue represents the ROI point cloud.

adaptive point cloud ROI extraction method based on shape extension.

The process of ROI extraction is illustrated in Fig. 8. Firstly, the original point cloud is appropriately downsampled to reduce point density while ensuring accuracy. Next, using the edge point cloud set as the seed point set and the original point cloud as the extension point set, expansion is performed using K Nearest Neighbors (KNN). To handle potential neighbor overlaps, a deduplication operation is performed on the resulting extended subset. Finally, the ROI point cloud is obtained.

The results of this adaptive ROI extraction are shown in Fig. 9. It is evident that this method not only reduces the introduction of noise points but also eliminates the cumbersome steps of removing background and irrelevant point clouds, streamlining the feature extraction process.

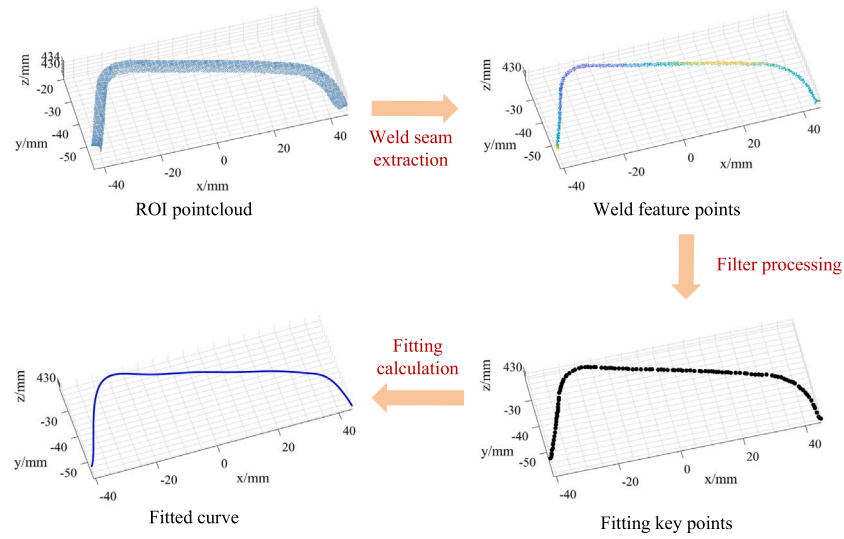


Fig. 10. Schematic diagram of the weld seam reconstruction process.

3.4. Weld seam reconstruction

As shown in Fig. 10, the geometric information contained within the extracted ROI point cloud is relatively simplistic. This simplicity allows us to further extract weld seam points using a method based on point cloud concave–convex features. Then, through density filtering, key points suitable for fitting are selected. Finally, employing a fitting algorithm, a smooth and accurate weld seam curve can be reconstructed.

3.4.1. Weld feature point extraction

Ref. [24] proposed a weld seam feature extraction method based on the Local Oriented Bounding Box (LOBB) point cloud descriptor. The researchers conducted an in-depth analysis of the role of the Principal Component Analysis (PCA) algorithm in spatial orientation and utilized the approximate geometric features of neighboring subsets to describe concave–convex features. Hierarchical feature extraction was then achieved using the K-means algorithm [39]. This method was experimentally validated on workpieces with various surface curvatures, demonstrating its accuracy and feasibility. This approach is chosen for weld seam extraction in this paper.

As illustrated in Fig. 11, the flatness in the LOBB feature descriptor characterizes the concavity and convexity of neighboring surfaces. Its principle is to describe the ratio of the height of the LOBB to the length, width, and height. In Fig. 11, we compare the LOBBs of the plane and crease points, respectively. When the surface exhibits significant concavity or convexity, the flatness value is higher; conversely, it is lower. We perform feature descriptions on the point cloud within the ROI, and the resulting feature heatmap is illustrated in Fig. 12. The red on the heatmap is primarily concentrated in the crease region, indicating that these points have a higher flatness value. This heatmap clearly demonstrates the advantages of the flatness feature, as it effectively highlights the features of the weld seam. Therefore, this method can be used to extract feature points of the weld seam.

As shown in Fig. 10, the extracted weld feature points essentially constitute a subset obtained from the point cloud data of the workpiece. These feature points are used to describe important information such as the morphology, position, and size of the weld. However, the characteristics of point cloud data are discrete and unordered, making it difficult to directly utilize feature points for spatial geometric descriptions of the weld. Therefore, based on the weld feature points, further data processing is required to achieve geometric spatial descriptions of the weld. This processing step is crucial for subsequent welding robot trajectory generation.

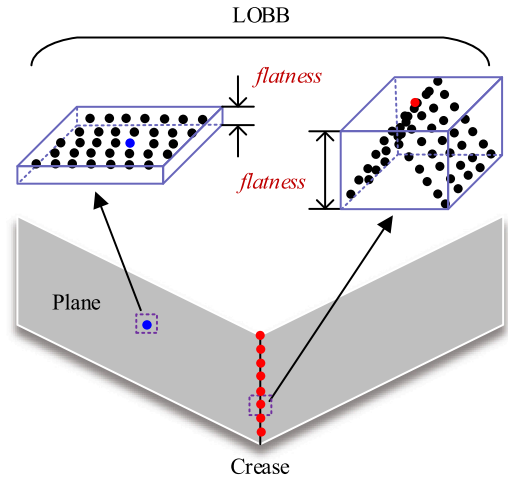


Fig. 11. Schematic diagram of LOBB feature description concavity and convexity principle. The blue, red, and black points represent the plane, crease, and neighboring points.

3.4.2. Fit key point filtering

Based on the projection principle of the DLP structured light camera, we know that the density of point cloud data obtained by the camera typically increases significantly at the creases of weld seams. This is because the creases of weld seams often exhibit more microstructures and geometric variations, resulting in more scattering and reflection of light spots in these areas. Therefore, in the point cloud data, the creases of weld seams often demonstrate higher data density and richer surface details.

Given this characteristic, we chose the Mean-Shift method to filter and extract key points from the point cloud of weld seam features, aiming to obtain the most representative fitting key points. The Mean-Shift method utilizes local density information of the data to achieve automatic clustering and feature point extraction, thereby effectively capturing key feature points at the creases of weld seams [40].

Here is the mathematical expression of the Mean-Shift algorithm on point clouds:

Assuming a point cloud dataset containing n points is represented as $X = \{x_1, x_2, x_3, \dots, x_n | x_i \in R^3\}$.

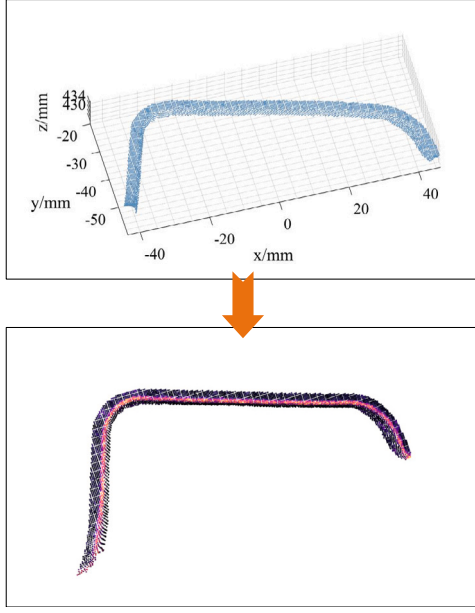


Fig. 12. Feature heatmap analysis. The redder the color, the larger the flatness value.

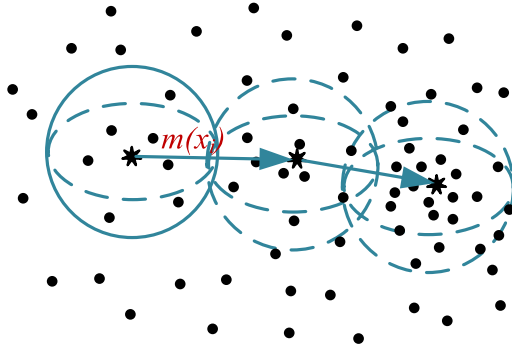


Fig. 13. Mean-Shift iterative processing. The arrows represent the iteration direction.

First, a kernel function $K(x)$ needs to be selected, commonly using the Gaussian kernel function:

$$K(x) = \frac{1}{(2\pi)^{d/2} h^d} \phi\left(\frac{\|x\|^2}{h^2}\right) \quad (4)$$

Where h denotes the bandwidth parameter, and $\phi(u)$ represents the Gaussian kernel function, defined as:

$$\phi(u) = e^{-\frac{u}{2}} \quad (5)$$

Next, for a given data point x_i , calculate its Mean-Shift vector $m(x_i)$:

$$m(x_i) = \frac{\sum_{x_j \in N(i)} K(x_j - x_i) x_j}{\sum_{x_j \in N(i)} K(x_j - x_i)} - x_i \quad (6)$$

Where $N(x_i)$ represents the set of neighboring points of x_i within a specified radius r , also referred to as the window.

As depicted in Fig. 13, the vector $m(x_i)$ always points in the direction of increased density in the point cloud. Based on this vector, the point's window sliding can be summarized into the iterative formula Eq. (7):

$$x_i^{t+1} = x_i^t + m(x_i^t) \quad (7)$$

The iterative process continues until the norm of the $m(x_i)$ falls below a specified threshold, at which point the algorithm is said to converge. After convergence, we can cluster the data points, typically by assigning each point to its nearest cluster center. Thus, the Mean-Shift algorithm continuously shifts data points toward the direction of

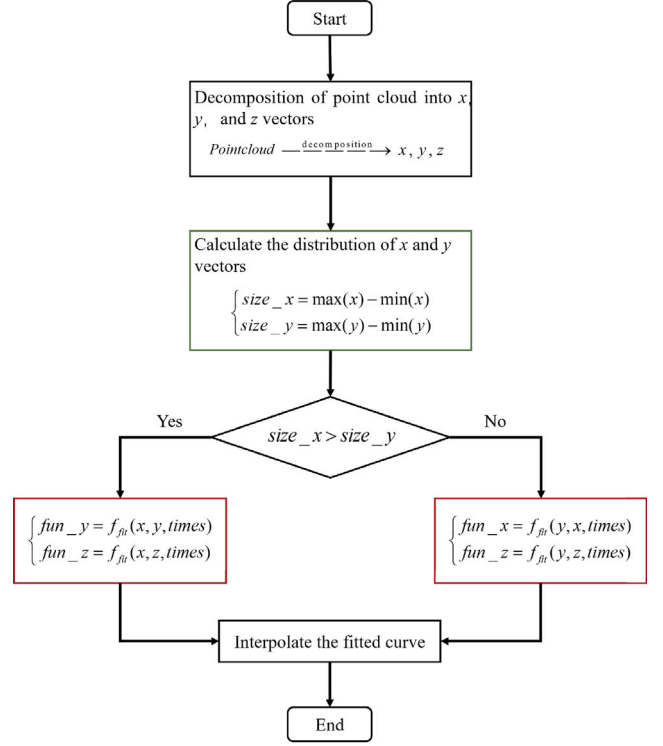


Fig. 14. The flowchart of the three-dimensional curve fitting algorithm based on the main axis distribution.

the highest local density, eventually aggregating them into regions of the highest density, thereby achieving clustering on the point cloud.

3.4.3. Curve fitting

To fit the key points, we employ a polynomial fitting method based on the least squares approach. The origin of the camera coordinate system is located at the optical center of the camera, with the x -axis pointing to the right of the camera and the y -axis pointing downward. When capturing the welding seam area, in order to capture rich details, the z -axis of the camera coordinate system is typically oriented as perpendicularly as possible to the welding seam area. Therefore, the point cloud obtained preliminarily in the camera coordinate system usually distributes primarily along the x -axis or y -axis.

Therefore, this paper proposes a three-dimensional curve fitting algorithm based on the main axis distribution to simplify the fitting model and reduce fitting errors. The processing procedure of this method is illustrated in Fig. 14. Firstly, vectors x , y , and z in three-dimensional space are extracted from the given point cloud data Pointcloud. Then, polynomial fitting is selectively performed based on the magnitude of changes in the curve along the x and y directions. Specifically, if the change magnitude along the x direction is greater than that along the y direction, fitting is performed in the x - y and x - z planes; conversely, if the change magnitude along the y direction is greater than that along the x direction, fitting is performed in the y - x and y - z planes. The term $f_{fit}(u, v, times)$ in the figure represents polynomial fitting, where u is the independent variable, v is the dependent variable, and $times$ is the order of the polynomial. The returned fun_v represents the optimal polynomial equation obtained, which is formulated as Eq. (8), where a_i are the coefficients of the polynomial.

$$fun_v = f_{fit}(u, v, times) = \sum_{i=0}^{times} a_i u^i \quad (8)$$

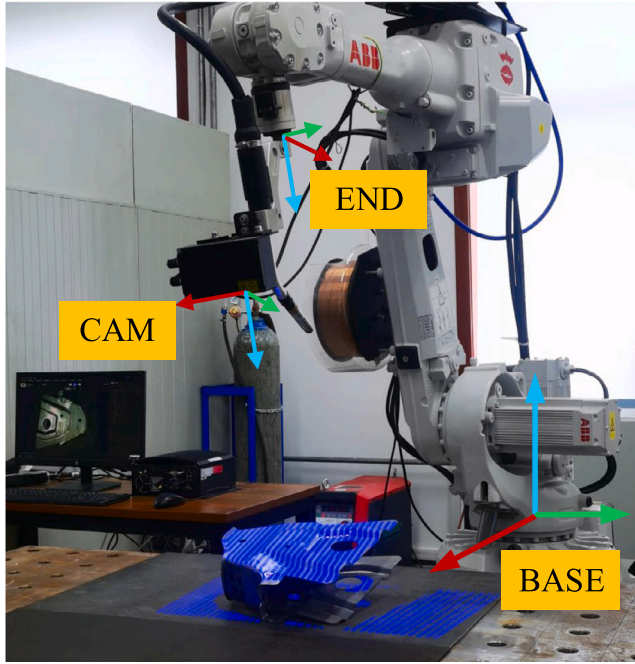


Fig. 15. Welding robot based on 3D vision experimental scene and coordinate systems.

4. Experiments

The experimental scene, as shown in Fig. 15, illustrates that the welding robot system, which employs three-dimensional vision, primarily involves three coordinate systems: the camera coordinate system (CAM), the robot end-effector coordinate system (END), and the robot base coordinate system (BASE). For ease of calculation, it is necessary to unify the system coordinate systems before conducting the experiment. The specific method involves determining the transformation matrix between CAM and END through hand-eye calibration, and then unifying the point cloud and trajectory to the BASE coordinate system based on robot kinematics [41]. This preparatory work has been completed prior to the experimental validation.

4.1. Weld seam coarse localization verification and analysis

To validate the performance of the K-Net neural network semantic segmentation model in segmenting ISWs, we conduct evaluations using a test dataset. Figs. 16(a), 16(b), and 16(c) depict the original images captured under different angles and lighting conditions, respectively. Figs. 16(d), 16(e), and 16(f) represent the ground-truth images, while Figs. 16(g), 16(h), and 16(i) illustrate the predicted images. Subsequently, we set the neighborhood size for edge extraction as 3, resulting in the edge extraction outcomes shown in Figs. 16(j), 16(k), and 16(l). The results demonstrate that the model is capable of robust and accurate segmentation of ISWs and coarse localization of weld seams.

To demonstrate the limitations of 2D edge extraction in direct weld reconstruction, we employ a curve fitted by manually accurate teaching points as the real weld curve. We select the roughly extracted edge point cloud of one of the welds and perform curve fitting on it as the reconstructed weld curve. As shown in Fig. 17, the two curves are uniformly interpolated to 80 points, and error analysis is performed on x , y , and z , respectively. By calculating the maximum errors of x , y , and z , they are 2.09 mm, 1.5 mm, and 2.26 mm, respectively. Additionally, due to the lack of depth, the z value errors of the extraction points are

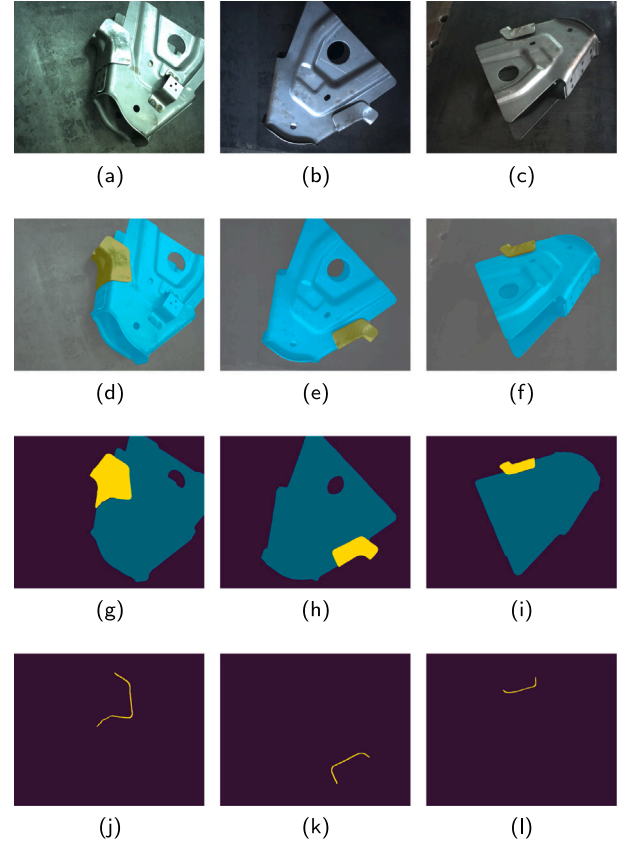


Fig. 16. Semantic segmentation and edge extraction results of the images. (a)–(c) Original images. (d)–(f) Ground-truth images. (g)–(i) Predicted images. (j)–(l) Edge extraction images.

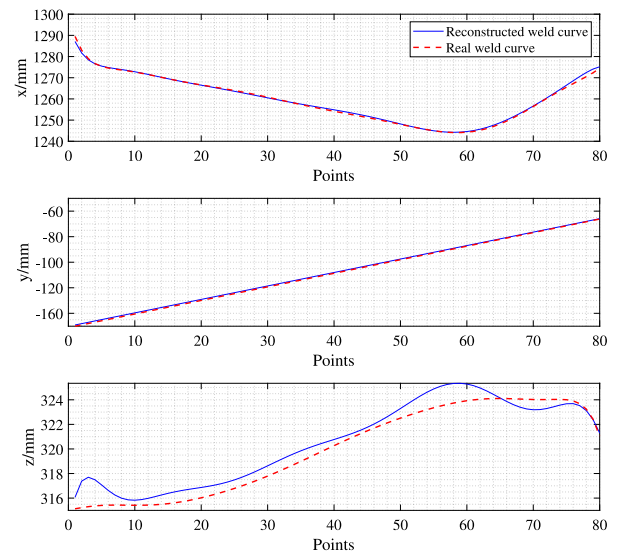


Fig. 17. Weld seam coarse localization error analysis chart.

very unstable. Therefore, the error does not meet the requirements of robot welding at all. This further proves the importance of extracting weld feature points based on the concave and convex features of the point cloud.

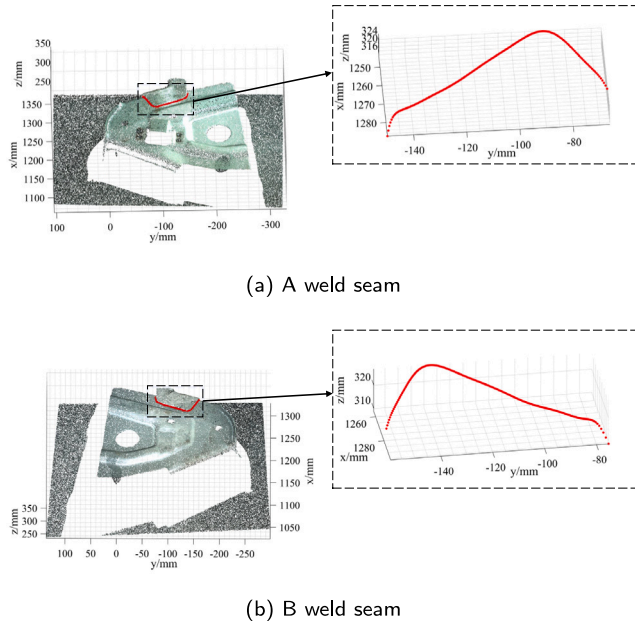


Fig. 18. Weld seam reconstruction results.

Table 2

Errors analysis.

Weld seam	Error	x (mm)	y (mm)	z (mm)
A	ME	1.0925	0.7972	0.8559
	RMSE	0.5682	0.4262	0.3499
B	ME	1.1270	0.8153	0.7436
	RMSE	0.6096	0.4718	0.2008

4.2. Weld reconstruction verification and analysis

According to the process requirements, the selected workpiece only needs to be welded on two sides by the robot. As illustrated in the experimental scene shown in Fig. 15, the 3D camera is capable of capturing only one weld seam at a time. Therefore, to capture the second weld seam, the workpiece needs to be manually or mechanically flipped to facilitate data collection for the other seam. The weld seam extracted and reconstructed by the algorithm proposed in this paper is shown in Fig. 18. For subsequent analysis, we refer to the weld seam shown in Fig. 18(a) as ‘A’ weld seam’ and the weld seam shown in Fig. 18(b) as ‘B weld seam.’

4.2.1. Accuracy analysis

To validate the accuracy of the method proposed, we continue to use precisely defined teaching points to fit smooth curves as the real weld seams and compare them with the reconstructed weld seams for error analysis. As shown in Fig. 19, after aligning the endpoints of the weld seam curves obtained by the two methods and interpolating them, there are 80 weld seam points. The errors in the x, y, and z directions of each set of weld seam points are analyzed separately. The weld seam errors obtained are shown in Table 2. It can be observed that the maximum error (ME) is all less than 1.2 mm, and the root mean square error (RMSE) is all less than 0.7 mm. This level of accuracy is entirely satisfactory for robotic welding requirements.

4.2.2. Efficiency analysis

An important indicator for robotic welding replacing manual labor is efficiency. Therefore, this paper conducts an analysis of the proposed method to demonstrate its efficiency. The industrial computer used in the welding robot platform built in this paper is equipped with an

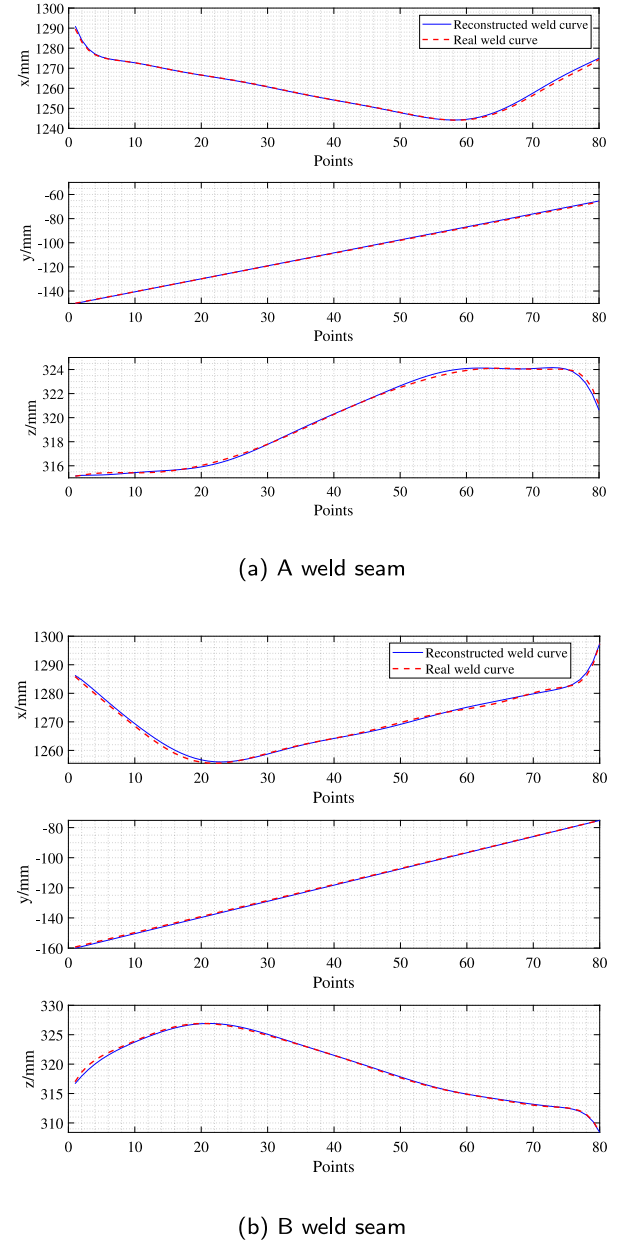


Fig. 19. Weld seam reconstruction accuracy analysis chart.

Table 3

Algorithm running time.

Step	Running time (ms)
RGB-D data collection	500
Semantic segmentation	1056
ROI extraction	204
Weld reconstruction	2630
The whole process	4390

8-core 2.3 GHz CPU and 16 GB of RAM. The algorithm is deployed and executed on this platform. Since the extraction process for weld seams A and B is the same, this paper runs the extraction process for A and B weld seams multiple times and finally calculates the average runtime for one weld seam, as shown in Table 3. It can be seen that the overall algorithm requires less than 5 s. This is a significant advantage compared to the traditional teach-and-playback mode.

4.3. Comparison with existing method

Given the complexity of ISW surfaces, current 3D vision-based weld seam extraction methods on the market often fall short. This has led to continued reliance on manual welding and teach-and-playback methods when dealing with such workpieces. To address this challenge, we propose a novel weld seam extraction method based on semantic segmentation and point cloud features, aiming to achieve robot welding without the need for programming or teaching. Experimental analysis has demonstrated that our proposed method can fully replace manual welding and teach-and-playback methods.

In previous research [24], we introduced a point cloud feature descriptor called LOBB, used for extracting edges and corners from workpiece point clouds. While this method is suitable for scenes with clearly defined weld seam convex and concave features, it is not applicable to surfaces as complex as ISWs. Therefore, this paper has made improvements based on this foundation. The proposed method utilizes neural networks to achieve workpiece segmentation and coarse localization of weld seams. Subsequently, we identify subsets of point clouds with prominent weld seam feature distributions and successfully extract weld seam features using the LOBB feature descriptor on these subsets. Compared to the original method [24], the proposed method in this paper exhibits greater adaptability. Additionally, the proposed method does not require simplification or preprocessing operations to remove backgrounds, making it more robust and reliable.

4.4. Discussion

The weld seam extraction method proposed in this study has demonstrated good performance in experiments. However, there are still some challenges in applying this framework to real-world welding scenarios.

1. Due to the diversity of robots used in different manufacturing environments, communication protocols and motion control methods vary. Therefore, designing flexible interfaces to achieve smoother system integration poses a significant challenge.
2. Real-world welding scenarios can be quite complex. To ensure that the robot safely avoids obstacles during the welding process, it is necessary to optimize the welding path planning in conjunction with a robot obstacle avoidance algorithm. This presents a challenge for improving the safety and efficiency of the welding process.
3. During welding, the proper adjustment of process parameters (such as speed, temperature, and welding sequence) is crucial for ensuring welding quality. How to dynamically adjust these parameters to accommodate the characteristics of different workpieces and welding requirements is also a challenge that needs to be addressed.

At the same time, this study still has several areas for improvement.

1. The proposed method has only been experimentally validated on the ISW corner joint weld seams. Future research could expand to include experiments on other types of weld joints (such as lap joints and V-shaped joints) to verify the generalizability and adaptability of the proposed method.
2. The current method shows certain robustness in weld feature extraction under different lighting conditions. However, when faced with scenes with large surface reflectivity, the depth information collected by the 3D camera may be partially missing. In this case, the robustness of the method may be challenged. Therefore, future research will focus on improving the method's robustness, especially in cases with significant changes in lighting and material.

5. Conclusion

This paper proposes an innovative method for extracting weld seam features, combining semantic segmentation and point cloud features, aimed at achieving robotic welding of irregular structure workpieces without teaching and programming. To address the research objective, we constructed a highly robust dataset first. By employing the K-Net semantic segmentation network for transfer learning, successful semantic segmentation of the workpiece was achieved. Subsequently, edge extraction was performed on the segmentation mask to achieve coarse localization of the weld seam. Then, using the principles of DLP structured light imaging, the edge point cloud reconstruction was conducted, and the weld seam ROI was extracted based on shape extension. Finally, based on the LOBB feature extraction method, weld seam feature points were successfully extracted, and the complete weld seam was reconstructed through fitting. The experimental results demonstrate that this method is both accurate, robust, and efficient in weld seam extraction.

CRedit authorship contribution statement

Yuankai Zhang: Writing – original draft, Software, Methodology, Investigation, Formal analysis, Conceptualization. **Yusen Geng:** Writing – original draft, Investigation, Data curation. **Xincheng Tian:** Writing – review & editing, Supervision, Resources, Funding acquisition, Conceptualization. **Yujie Sun:** Formal analysis, Data curation. **Xiaolong Xu:** Formal analysis, Data curation.

Declaration of competing interest

The authors that they do not have any commercial or associative interest that represents a conflict of interest in connection with the work submitted.

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Data availability

Data will be made available on request.

References

- [1] W. Fang, X. Tian, A novel model-based welding trajectory planning method for identical structural workpieces, *Robot. Comput.-Integr. Manuf.* 89 (2024) 102772.
- [2] S. Wang, Z. Xu, C. Wu, L. Hua, D. Zhu, Towards region-based robotic machining system from perspective of intelligent manufacturing: A technology framework with case study, *J. Manuf. Syst.* 70 (2023) 451–463.
- [3] C. Zheng, Y. An, Z. Wang, H. Wu, X. Qin, B. Eynard, Y. Zhang, Hybrid offline programming method for robotic welding systems, *Robot. Comput.-Integr. Manuf.* 73 (2022) 102238.
- [4] Y. Zou, R. Zhan, Research on 3D curved weld seam trajectory position and orientation detection method, *Opt. Lasers Eng.* 162 (2023) 107435.
- [5] H. Wang, C. Chen, Y. Liu, B. Ren, Y. Zhang, X. Zhao, Y. Chi, A novel approach for robotic welding trajectory recognition based on pseudo-binocular stereo vision, *Opt. Laser Technol.* 174 (2024) 110669.
- [6] Z. Hou, Y. Xu, R. Xiao, S. Chen, A teaching-free welding method based on laser visual sensing system in robotic GMAW, *Int. J. Adv. Manuf. Technol.* 109 (5) (2020) 1755–1774.
- [7] T. Lei, Y. Rong, H. Wang, Y. Huang, M. Li, A review of vision-aided robotic welding, *Comput. Ind.* 123 (2020) 103326.

- [8] Y. He, G. Ma, S. Chen, Autonomous decision-making of welding position during multipass GMAW with T-joints: A Bayesian network approach, *IEEE Trans. Ind. Electron.* 69 (4) (2021) 3909–3917.
- [9] A. Rout, B. Deepak, B.B. Biswal, G.B. Mahanta, Weld seam detection, finding, and setting of process parameters for varying weld gap by the utilization of laser and vision sensor in robotic arc welding, *IEEE Trans. Ind. Electron.* 69 (1) (2021) 622–632.
- [10] Y. Zou, X. Chen, G. Gong, J. Li, A seam tracking system based on a laser vision sensor, *Measurement* 127 (2018) 489–500.
- [11] J. Fan, S. Deng, F. Jing, C. Zhou, L. Yang, T. Long, M. Tan, An initial point alignment and seam-tracking system for narrow weld, *IEEE Trans. Ind. Inform.* 16 (2) (2019) 877–886.
- [12] J. Cai, T. Lei, An autonomous positioning method of tube-to-tubesheet welding robot based on coordinate transformation and template matching, *IEEE Robot. Autom. Lett.* 6 (2) (2021) 787–794.
- [13] X. Li, X. Li, S.S. Ge, M.O. Khyam, C. Luo, Automatic welding seam tracking and identification, *IEEE Trans. Ind. Electron.* 64 (9) (2017) 7261–7271.
- [14] Y. Ma, J. Fan, S. Deng, Y. Luo, X. Ma, F. Jing, M. Tan, Efficient and accurate start point guiding and seam tracking method for curve weld based on structure light, *IEEE Trans. Instrum. Meas.* 70 (2021) 1–10.
- [15] T. Lei, Y. Huang, W. Shao, W. Liu, Y. Rong, A tactual weld seam tracking method in super narrow gap of thick plates, *Robot. Comput.-Integr. Manuf.* 62 (2020) 101864.
- [16] Y. Zou, T. Chen, Laser vision seam tracking system based on image processing and continuous convolution operator tracker, *Opt. Lasers Eng.* 105 (2018) 141–149.
- [17] C. Lyu, P. Li, D. Wang, S. Yang, Y. Lai, C. Sui, High-speed optical 3D measurement sensor for industrial application, *IEEE Sensors J.* 21 (10) (2020) 11253–11261.
- [18] W. Guo, X. Huang, B. Qi, X. Ren, H. Chen, X. Chen, A novel versatile method for generating machining path directly from point cloud, *J. Manuf. Process.* 120 (2024) 15–27.
- [19] Y. Geng, Y. Zhang, X. Tian, L. Zhou, A novel 3D vision-based robotic welding path extraction method for complex intersection curves, *Robot. Comput.-Integr. Manuf.* 87 (2024) 102702.
- [20] Y. Geng, M. Lai, X. Tian, X. Xu, Y. Jiang, Y. Zhang, A novel seam extraction and path planning method for robotic welding of medium-thickness plate structural parts based on 3D vision, *Robot. Comput.-Integr. Manuf.* 79 (2023) 102433.
- [21] Z. Lu, J. Fan, Z. Hou, S. Deng, C. Zhou, F. Jing, Automatic 3D seam extraction method for welding robot based on monocular structured light, *IEEE Sensors J.* 21 (14) (2021) 16359–16370.
- [22] Z. Yuankai, J. Yong, T. Xincheng, X. Xiaolong, G. Yusen, L. Min, A point cloud-based welding trajectory planning method for plane welds, *Int. J. Adv. Manuf. Technol.* 125 (3) (2023) 1645–1659.
- [23] F. Xu, Z. Hou, R. Xiao, Y. Xu, Q. Wang, H. Zhang, A novel welding path generation method for robotic multi-layer multi-pass welding based on weld seam feature point, *Measurement* 216 (2023) 112910.
- [24] Y. Zhang, Y. Geng, X. Tian, F. Zheng, Y. Jiang, M. Lai, A feature extraction approach over workpiece point clouds for robotic welding, *IEEE Trans. Autom. Sci. Eng.* (2024).
- [25] P. Zhou, R. Peng, M. Xu, V. Wu, D. Navarro-Alarcon, Path planning with automatic seam extraction over point cloud models for robotic arc welding, *IEEE Robot. Autom. Lett.* 6 (3) (2021) 5002–5009.
- [26] S.M. Ahmed, Y.Z. Tan, C.M. Chew, A. Al Mamun, F.S. Wong, Edge and corner detection for unorganized 3D point clouds with application to robotic welding, in: 2018 IEEE/RSJ International Conference on Intelligent Robots and Systems, IROS, IEEE, 2018, pp. 7350–7355.
- [27] H. Wang, Y. Huang, G. Zhang, Y. Rong, A novel method for dense point cloud reconstruction and weld seam detection for tubesheet welding robot, *Opt. Laser Technol.* 163 (2023) 109346.
- [28] V. Patil, I. Patil, V. Kalaichelvi, R. Karthikeyan, Extraction of weld seam in 3D point clouds for real time welding using 5 dof robotic arm, in: 2019 5th International Conference on Control, Automation and Robotics, ICCAR, IEEE, 2019, pp. 727–733.
- [29] J. Gao, F. Li, C. Zhang, W. He, J. He, X. Chen, A method of d-type weld seam extraction based on point clouds, *IEEE Access* 9 (2021) 65401–65410.
- [30] W. Guo, X. Huang, B. Qi, X. Ren, H. Chen, X. Chen, Vision-guided path planning and joint configuration optimization for robot grinding of spatial surface weld beads via point cloud, *Adv. Eng. Informatics* 61 (2024) 102465.
- [31] Y. Ma, J. Fan, S. Zhao, F. Jing, S. Wang, M. Tan, From model to reality: A robust framework for automatic generation of welding paths, *IEEE Trans. Ind. Electron.* (2024).
- [32] T. Wan, S. Du, W. Cui, R. Yao, Y. Ge, C. Li, Y. Gao, N. Zheng, RGB-D point cloud registration based on salient object detection, *IEEE Trans. Neural Netw. Learn. Syst.* 33 (8) (2021) 3547–3559.
- [33] W. Zhang, J. Pang, K. Chen, C.C. Loy, K-net: Towards unified image segmentation, *Adv. Neural Inf. Process. Syst.* 34 (2021) 10326–10338.
- [34] T. Xiao, Y. Liu, B. Zhou, Y. Jiang, J. Sun, Unified perceptual parsing for scene understanding, in: Proceedings of the European Conference on Computer Vision, ECCV, 2018, pp. 418–434.
- [35] Z. Liu, Y. Lin, Y. Cao, H. Hu, Y. Wei, Z. Zhang, S. Lin, B. Guo, Swin transformer: Hierarchical vision transformer using shifted windows, in: Proceedings of the IEEE/CVF International Conference on Computer Vision, 2021, pp. 10012–10022.
- [36] MMSegmentation Contributors, MMSegmentation: OpenMMLab semantic segmentation toolbox and benchmark, 2020, <https://github.com/open-mmlab/mms Segmentation>.
- [37] B. Zhou, H. Zhao, X. Puig, S. Fidler, A. Barriuso, A. Torralba, Scene parsing through ADE20K dataset, in: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, 2017, pp. 633–641.
- [38] S. Zhang, High-speed 3D shape measurement with structured light methods: A review, *Opt. Lasers Eng.* 106 (2018) 119–131.
- [39] F. Nie, H. Liu, R. Wang, X. Li, Parameter-free multiview *K*-means clustering with coordinate descent method, *IEEE Trans. Neural Netw. Learn. Syst.* (2024).
- [40] Z. Hui, S. Jin, Y. Xia, Y. Nie, X. Xie, N. Li, A mean shift segmentation morphological filter for airborne LiDAR DTM extraction under forest canopy, *Opt. Laser Technol.* 136 (2021) 106728.
- [41] Y. Song, W. Tian, Y. Tian, X. Liu, An efficient calibration method for serial industrial robots based on kinematics decomposition and equivalent systems, *Robot. Comput.-Integr. Manuf.* 84 (2023) 102607.